COURSE CODE / NAME: AA 200 BIOMETRY

Background and Rationale

The aim of the course is to provide students with an understanding of the statistical methods and data analysis techniques used in research so as to enable students to apply such methods and techniques in the field of biological sciences.

Learning Outcomes

At the completion of the course students should be able to understand:

- Define relevant statistical terminologies, organize and present data using frequency distributions
- Construct and interpret graphical displays such as histograms, box and whisker plots, bar charts, pie charts, stem and leaf and scatter plots
- Calculate and interpret summary statistics for data sets
- Understand and explain what sampling method (s) to use in a given situation
- Recognize and understand the relevance of probability theory and probability distributions, such as Binomial, Poisson and Normal, in analysis of biological data
- Understand the difference and link between descriptive and inferential statistics
- Construct and correctly interpret confidence intervals around point estimates
- Formulate and conduct statistical hypothesis testing
- Recognize the situation when it is appropriate to use and be able to perform: two sample (independent and paired) *t*-tests, one-way and two-way ANOVA, correlation and simple linear regression analysis and r x c contingency table analysis
- Know the conditions (or assumptions) required for the validity of the stated statistical tests, and know what alternatives are available if the assumptions are not met
- Understand the proper use and interpretation of significance levels (p-values)
- Know how to use statistical tables (especially *t*, *f*,χ^2 and standard normal Tables) and statistical software such as GenStat, R and SPSS to describe and analyze data and interpret outputs
- Know how to write a concise explanation of statistical results

Course Content

1.0 Definition of Basic Statistical Terminologies

- Definition of terms: Statistics, Biometry, Descriptive and Inferential Statistics, population, sample, parameter, statistic, variable, data, parametric and non-parametric statistics

- Types of variables and data
- Scales of measurement (Nominal, Ordinal, Interval and Ratio Scales)

2.0 Introduction to Elementary Statistics

- Organizing and presenting data: frequency distributions and graphs (histograms, frequency polygons, ogive, dot plots, stem-and-leaf plots, pie charts and bar charts)
- Measures of central tendency (mean, median and mode)
- Measures of variability or dispersion (range, interquartile range, variance and standard deviation, coefficient of variation)
- Empirical rule
- Measures of shape of a distribution (Skewness and Kurtosis)
- Measures of position (standard scores, percentiles, quartiles, outliers)
- Exploratory Data Analysis (five-number summary, box-and-whiskers plot)

3.0 Introduction to Sampling Methods

- Definition of terms: census, survey, population, sample, sampling, target and sampled population, sampling unit, observation unit, sampling frame, sampling method, probability sampling method, non-probability sampling method
- Probability sampling methods: simple random sampling, systematic, stratified and cluster sampling
- Non-probability sampling methods: convenience, purposive, snowball and quota sampling

4.0 Elementary Probability Theory

- Definitions: random experiment, trial, event, outcome, sample space, probability
- Basic event relations and probability laws: intersection, union, compliment, mutually exclusive, conditional probability and independent events
- Counting rules (fundamental counting rule, permutations and combinations) and their use in probability
- Bayes' Theorem

5.0 Introduction to Probability Distributions

- Definition of terms: random variable, discrete and continuous random variables, probability distribution, expected value (mean) and standard deviation for a random variable or probability distribution
- Computation of mean and standard deviation for a discrete and continuous probability distribution

- Examples of discrete probability distributions (Bernoulli, Binomial and Poisson)
- Examples of continuous probability distributions (Normal, t, χ^2, f)

6.0 Sampling Distribution

- Definition of sampling distribution
- Sampling distribution of the sample mean
- Sampling distribution of the sample proportion
- The central limit theorem
- Sampling from a normal population

7.0 Introduction to Statistical Inference

- Definition of terms: estimation, point and interval estimation and hypothesis testing
- Properties of a good estimator
- Inferences (estimation and hypothesis testing) involving the mean (i.e. one and two samples tests): known and unknown population standard deviation; including independent and paired samples.
- Inferences involving the proportion: one and two samples tests
- Inferences involving the variance and standard deviation: one and two samples tests.

8.0 Principles of Experimental Design and Analysis of Variance (ANOVA)

- Definition of concepts: experiment, design, experimental design, experimental unit, measurement unit, factor, treatment, experimental error, sums of squares, variance, ANOVA
- Basic principles of experimental design: randomization, replication and blocking
- One-way ANOVA (Completely Randomized Design [CRD])
- Two-way ANOVA (Randomized Complete Block Design [RCBD])
- Multiple Comparison Tests (e.g. Least Significant Difference, Dunnett and Tukey Methods)
- Latin square design(design features, linear statistical model and partitioning of sums of squares)
- Factorial experiments (treatment structure, interactions, factorial experiment in CRD and in RCBD)
- Split-plot design(design features, linear statistical model and partitioning of sums of squares)
- Handling Violations of ANOVA assumptions; Data Transformation (e.g. Logarithmic,

Square Root and Arcsine)

9.0 Introduction to Non-Parametric Statistics

- Introduction
- Non-parametric alternatives to t-tests (Sign Test, Wilcoxon Rank Sum Test/Mann-Whitney U-Test and Wilcoxon Signed-Rank Test)
- Non-parametric alternatives to CRD (Kruskal-Wallis Test) and RCBD (Friedman Test)

10.0 Introduction to Bivariate Data Analysis

- Introduction to correlation and regression analysis (simple linear regression)
- Ordinary Least Squares (OLS) estimation of regression parameters
- ANOVA approach to regression analysis
- Introduction to multiple linear regression analysis

11.0 Analysis of Categorical Data

- Introduction to analysis of categorical data
- Chi-square test for goodness-of-fit
- Contingency tables: chi-square tests for independence and homogeneity

12.0 Introduction to Statistical Packages

- Introduction to GenStat/R/SPSS
- Descriptive statistics using GenStat/R/SPSS
- One and two samples t-tests using GenStat/R/SPSS
- One and two-way ANOVA using GenStat/R/SPSS
- Checking ANOVA assumptions and data transformation using GenStat/R/SPSS
- Non-parametric statistics using GenStat/R/SPSS
- Correlation and regression analysis using GenStat/R/SPSS
- Categorical data analysis using GenStat/R/SPSS

Assessment

- Continuous Assessment 40%

  Final Exam 60%

Recommended References

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3. Field, A., Miles, J. and Field, Z. (2012). Discovering Statistics Using R. SAGE Publications Ltd, London, Great Britain. ISBN 978-1-4462-0045-2
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12. Zar, J.H. (2014). Biostatistical Analysis, 5th Edition; Pearson Education Limited, England, USA. ISBN 10:1-292-02404-6, ISBN 13:978-1-292-02404-2